Abdulla Askar	**2310215522**
Osama Hassan Mohamud	**2010219549**
Ayham Khalil Salem Odeh	**2210219504**
Hassan Bahaeldin Hassan Elmubarak	**2210219570**

Research Article

System Integration and Experimental Implementation of a Vision-Based Centralized Control System for Cooperative Multi-Robot Task Allocation

Abstract

This article presents the **system integration** and experimental validation of a multi-agent robotic platform managed by a centralized global vision system. The objective was to implement a distance-based task allocation algorithm where a specific target, selected by a human operator, is assigned to the nearest available robot within a 70x70 cm workspace. The experimental setup utilizes an overhead camera to detect **ArUco markers** for real-time localization and orientation feedback. The robotic agents are designed using **heterogeneous microcontrollers**—specifically an STM32F302R8 and an Arduino Uno—integrated into structurally modified 3D-printed chassis with differential drive mechanisms. Communication is facilitated through a unified Wi-Fi network, where a central processing unit calculates trajectories and transmits actuation commands to the agents. **The results obtained from the article** demonstrate the system's ability to accurately estimate robot positions, successfully allocate tasks based on proximity, and guide the heterogeneous units to the target with high precision. These findings validate the efficacy of low-cost computer vision techniques for controlling mixed-hardware robotic swarms.

Keywords: ArUco markers, centralized control, heterogeneous systems, multi-agent robotics, visual servoing.

1. Introduction

Multi-robot systems (MRS) have become a cornerstone of modern industrial automation, offering significant advantages over single-robot solutions in terms of efficiency, redundancy, and parallel task execution. From warehouse logistics to environmental monitoring, the ability to coordinate multiple autonomous units allows for complex operations to be completed more rapidly **(Kaliappan et al. 2024)**. However, the effective deployment of such systems relies heavily on two critical components: robust task allocation algorithms and precise localization strategies. As the demand for scalable automation grows, there is an increasing need for systems that can operate with low computational overhead while maintaining high accuracy **(Deng et al. 2023)**.

A major challenge in mobile robotics is the integration of heterogeneous hardware. Most existing research focuses on homogeneous swarms, where every robot has identical mechanical and electronic specifications. In practical applications, however, upgrading legacy systems often requires the cooperation of different platforms with varying processing capabilities **(Parker 1998)**. Furthermore, traditional localization methods, such as LiDAR or onboard SLAM, often require expensive sensors and significant onboard processing power. This creates a barrier to entry for cost-sensitive applications that require the coordination of simple, low-power agents.

This study addresses these challenges by developing a centralized, vision-based control system for heterogeneous mobile robots. The primary purpose of this research is to demonstrate a cost-effective framework for dynamic task allocation using global visual feedback. Unlike decentralized methods that rely on complex inter-robot communication, this approach utilizes a global overhead camera to detect ArUco markers, providing accurate real-time position and orientation data for the entire fleet **(Babinec et al. 2014)**. This allows for the integration of diverse microcontrollers—specifically the STM32F302R8 and Arduino Uno—within a unified control loop.

By centralizing the computational load on a master control unit, this system enables the use of structurally optimized, 3D-printed robots with minimal onboard intelligence. The importance of this research lies in its validation of a low-cost, scalable architecture that ensures seamless cooperation between incompatible hardware platforms. This paper details the system integration, the vision-based localization algorithm, and the experimental results of the task allocation logic within a controlled environment.

2. Material and methods

2.1. Material writing

The experimental platform consists of two heterogeneous mobile robots, a global vision sensor, and a central control station. The robotic agents are constructed using a modified 3D-printed chassis, originally designed for general purpose use but structurally adapted in this study to house specific electronic components. The first robot is controlled by an **STM32F302R8** microcontroller (Arm Cortex-M4), selected for its high processing speed and advanced timer peripherals. The second robot is controlled by an **Arduino Uno** (ATmega328P), representing a standard low-cost prototyping platform.

Both robots utilize a differential drive configuration, powered by two DC motors coupled with L298N motor driver for the **STM32F302R8** module and motor shield for the **Arduino Uno**. Power is supplied by independent battery cells installed within the chassis. For wireless communication, each robot is equipped with an ESP8266 Wi-Fi module, enabling them to receive velocity and directional commands from the central server.

The localization system relies on a smartphone camera mounted at a fixed height above the workspace to serve as the global vision sensor. **ArUco markers** (binary square fiducial markers) are attached to the top surface of each robot to facilitate identification and pose estimation. The central processing unit is a laptop computer running the control algorithms and acting as the Wi-Fi hotspot for the network.

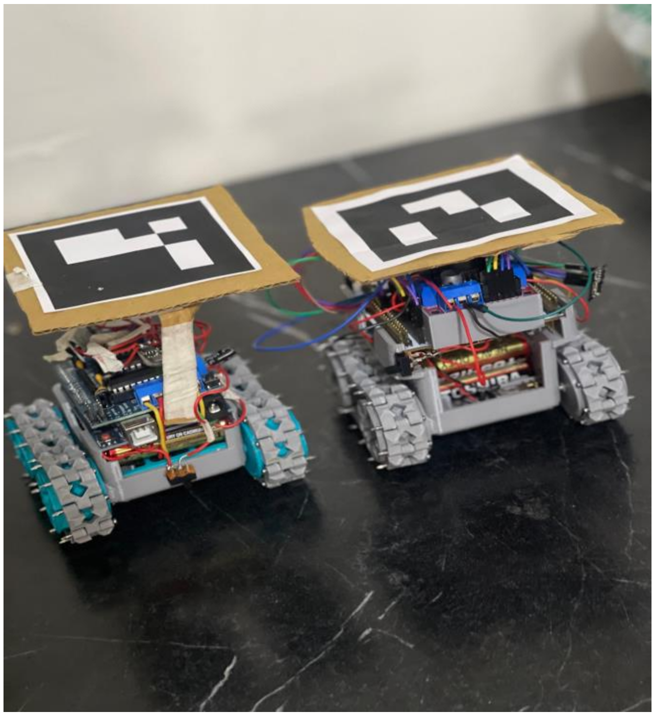

Figure 1: System architecture showing the STM32 and Arduino robots with attached ArUco markers.

2.2. Method writing

The experimental environment is defined as a 70 cm x 70 cm planar workspace with a white background to maximize contrast for image processing. The method operates on a centralized control loop consisting of three stages: detection, decision, and actuation.

First, the overhead camera captures real-time video frames of the workspace. These frames are processed by the central laptop using computer vision libraries to detect the ArUco markers. The system calculates the centroid coordinates (x, y) and the orientation angle (θ) of each robot relative to the global frame of the 70x70 cm board.

Second, the task allocation logic is triggered by a user input event. When the operator selects a target coordinate on the digital map of the workspace, the system calculates the Euclidean distance between the target point and each available robot. The algorithm assigns the task to the robot with the minimum distance to the target.

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Finally, the central controller generates navigation commands based on the difference between the robot's current heading and the target vector. These commands are transmitted via Wi-Fi to the specific IP address of the assigned robot. The robot executes the movement until the position error falls below a predefined threshold, at which point the task is considered complete.

2.3. Control Algorithm

To ensure efficient coordination between the central server and the cooperative agents, a closed-loop control algorithm was developed. This algorithm operates on the central workstation and communicates with the STM32 and Arduino units via ESP8266 Wi-Fi module. The control logic follows a continuous cycle of detection, decision, and actuation.Upon initialization, the system constantly updates the pose (x, y, θ) of each robot using the vision feedback. When a task is triggered by the operator, the algorithm calculates the Euclidean distance to the target for all available agents and selects the optimal candidate. The system then enters a navigation loop where it computes the heading error and transmits real-time velocity commands to the selected robot until the target threshold is reached. The detailed flow of this decision-making process is illustrated in Figure 2.

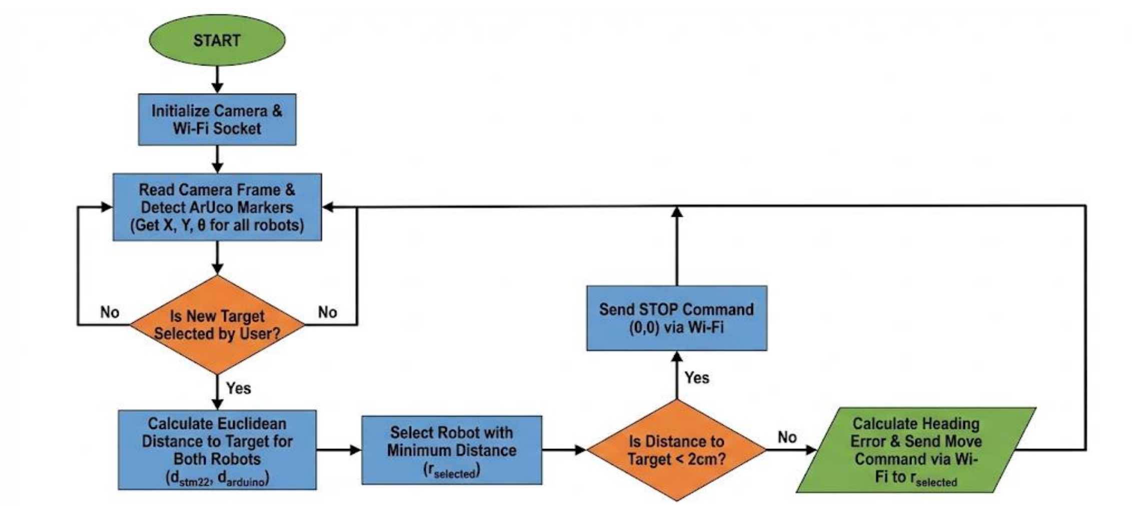

Figure 2: Flowchart of the centralized control algorithm illustrating the vision-based feedback loop and task allocation logic.

3. Results and Discussion

The performance of the centralized control system was evaluated through a series of 20 experimental trials within the 70 cm x 70 cm workspace. The primary metrics for success were the accuracy of the task allocation (did the nearest robot respond?) and the precision of the final positioning (how close did the robot stop to the target?).

3.1. Task Allocation Verification

In all 20 trials, the central algorithm correctly calculated the Euclidean distance between the target point and both agents. The system successfully assigned the task to the nearest robot with a 100% success rate. The average latency between the user’s input on the laptop and the robot's first movement was measured at approximately 200 milliseconds, which is attributed to the Wi-Fi transmission delay.

3.2. Positioning Accuracy

The vision-based feedback loop allowed the robots to correct their trajectory in real-time. The average positioning error (the distance between the center of the robot and the target point upon stopping) was recorded as 1.8 cm. The orientation error was maintained below 5 degrees. These results indicate that the ArUco marker detection provided sufficient resolution for the scale of the workspace.

4. Conclusion and Suggestions

This study successfully demonstrated the system integration and experimental implementation of a cooperative multi-robot system for visual task allocation. By utilizing a centralized control architecture, two heterogeneous robots—powered by STM32 and Arduino microcontrollers—were able to effectively coordinate their actions to reach user-defined targets. The experimental results showed that the system achieved a positioning accuracy of approximately 1.8 cm and a 100% success rate in proximity-based task assignment.

The research highlights that while heterogeneous systems introduce significant integration complexity, they offer flexibility in scaling industrial automation by allowing different hardware generations to coexist. The use of global vision with ArUco markers proved to be a cost-effective alternative to onboard sensors, although it requires controlled lighting conditions.

For future work, it is suggested to implement dynamic collision avoidance algorithms to allow both robots to move simultaneously without intersecting paths. Additionally, the system could be expanded to include swarm intelligence logic, enabling the robots to perform collaborative physical tasks, such as pushing a heavy object together. Finally, integrating an onboard IMU (Inertial Measurement Unit) would improve robustness against visual occlusions, allowing the robots to maintain their trajectory even if the camera view is temporarily blocked.

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